

A Random-Effects Model Reveals Strong Positive Sorting in CEO Labor Markets

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If markets allocate CEOs efficiently across firms, better managers should sort into better firms. The correlation between firm and manager quality is therefore central to understanding misallocation and aggregate productivity. Because manager quality is unobserved, the standard empirical strategy from matched worker–firm data is to estimate latent firm and worker effects and ask whether higher-quality workers sort to higher-quality firms. In CEO labor markets, however, careers are short and mobility is sparse, so fixed-effects estimates of latent quality are noisy and their implied correlation is badly biased. We instead model firm and manager effects as a Gaussian Markov random field on the bipartite CEO–firm network. Estimating four distributional parameters—rather than hundreds of thousands of individual effects—avoids limited mobility bias, while the sparsity of the precision matrix makes likelihood-based estimation feasible on the full network. Applied to Hungarian administrative data from 1990 to 2018, the model yields strong positive assortative matching ($\rho = 0.7$). By contrast, two-way fixed effects on the same data imply $\rho \approx -0.6$, and leave-one-out bias correction reduces the magnitude but does not resolve the discrepancy. In a model-based counterfactual, perfect sorting ($\rho = 1$) would raise aggregate output by about 6%.

assortative matching | CEO–firm sorting | Gaussian Markov random field | bipartite network | random effects | limited mobility bias

The impact of individual CEOs on firm performance is a central question in economics and corporate finance. If better CEOs systematically work at better firms, complementarities in production amplify output; if matching is inefficient, the economy forgoes gains from improved allocation. Measuring CEO–firm sorting is therefore important for understanding misallocation and aggregate productivity (1).

The empirical challenge is that CEO quality is unobserved. The standard empirical framework comes from the broader worker–firm matched-panel literature, which estimates firm and worker effects and studies their covariance to infer sorting (2, 3). This framework is natural to apply to CEOs because they are workers matched to firms, but CEO labor markets are a particularly difficult case: careers are short, firm transitions are rare, and the mobility network is thin. As a result, individual fixed effects remain noisy even in very large samples, which can severely distort estimated variance components and implied sorting patterns (4, 5, 6). Existing corrections—including parametric adjustments, leave-one-out procedures, and grouped heterogeneity models (3, 4, 6, 15, 16)—were developed largely in worker–firm settings and become especially fragile in sparse CEO data, a point also emphasized in recent work on CEO effects (7).

We develop a random-effects model, where firm and manager effects are drawn from a joint distribution on the bipartite

mobility network. The model has four parameters: the variance of firm effects σ_a^2 , the variance of manager effects σ_z^2 , their correlation ρ , and a match-specific noise variance σ_ε^2 . This formulation replaces hundreds of thousands of latent effects with a small set of distributional parameters, so it avoids limited-mobility bias and uses information from the full network, including disconnected components.

The specification induces a Gaussian Markov random field (GMRF) on the CEO–firm graph (8). That representation is useful for both identification and computation. Identification comes from correlation decay with network distance: along longer paths, covariances are attenuated by additional powers of ρ , echoing Clark’s (9) use of surname correlation decay to study intergenerational mobility. Computation comes from sparsity: the GMRF precision matrix has nonzero entries only along network links, which makes likelihood-based estimation feasible without inverting dense covariance matrices.

We apply the model to Hungarian administrative data covering firms and their CEOs from 1990 to 2018. We find strong positive sorting. In our baseline sample, which keeps connected components with at least 2 firms and 2 managers, the estimated correlation is $\rho = 0.71$; on the giant component, it rises to about 0.75. Standard two-way fixed effects applied to the same data imply instead a correlation of about -0.6 , and leave-one-out correction reduces the magnitude but does not remove the discrepancy. In the estimated variance decomposition, the firm–manager covariance accounts for about 6% of total revenue variance. A counterfactual with perfect sorting ($\rho = 1$) raises aggregate output by about 6%.

Our contribution is to show how assortative matching can be identified and estimated in extremely sparse bipartite mobility networks. The approach builds on the matched-effects tradition in labor economics (2, 3), connects naturally to the GMRF literature in spatial statistics (8), and provides evidence of strong positive CEO–firm sorting that complements the broader literature on managerial talent, CEO transitions, and misallocation (1, 10, 11, 12, 13).

Model

Setup. Consider a bipartite network of firms $i \in \mathcal{I}$ and managers $m \in \mathcal{J}$, with $N_f = |\mathcal{I}|$ firms and $N_m = |\mathcal{J}|$ managers. An employment spell between firm i and manager m is an edge in the graph. Let K denote the total number of observed spells. For each spell, we observe an outcome y_{im} , such as the log revenue of firm i while manager m is in office. We decompose that outcome as

$$y_{im} = a_i + z_m + \varepsilon_{im}, \quad [1]$$

where a_i is a firm effect, z_m is a manager effect, and ε_{im} is a match-specific disturbance. The disturbance ε_{im} is mean-zero,

independent across matches, and independent of (a_i, z_m) , with variance σ_ε^2 .

We treat firm and manager effects as random draws from a joint distribution. This is the key departure from the Abowd-Kramarz-Margolis approach (2), which treats these effects as fixed parameters to be estimated one by one.

The GMRF specification. Let $\mathbf{x} = (a_1, \dots, a_{N_f}, z_1, \dots, z_{N_m})^\top$ denote the stacked vector of all firm and manager effects. We model $\mathbf{x}$ as a Gaussian Markov random field on the bipartite graph with precision matrix

$$\mathbf{Q} = \mathbf{S}^{-1}(\mathbf{D} - \rho \mathbf{A})\mathbf{S}^{-1}, \quad [2]$$

where $\mathbf{A}$ is the bipartite adjacency matrix

$$\mathbf{A} = \begin{pmatrix} \mathbf{0}_{N_f \times N_f} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{0}_{N_m \times N_m} \end{pmatrix},$$

$\mathbf{B}$ is the $N_f \times N_m$ incidence matrix with $B_{im} = 1$ if firm i employed manager m , $\mathbf{D}$ is the diagonal degree matrix with $D_{kk} = \sum_\ell A_{k\ell}$, and $\mathbf{S} = \text{diag}(\sigma_a \mathbf{1}_{N_f}, \sigma_z \mathbf{1}_{N_m})$ is the variance scaling matrix. The parameter $\rho \in (-1, 1)$ governs the strength and sign of dependence between connected firm and manager effects, while σ_a and σ_z scale firm and manager heterogeneity.

The precision matrix $\mathbf{Q}$ is a variance-scaled graph Laplacian modified by the correlation parameter. Its sparsity is central: entries are nonzero only when two nodes are directly linked in the bipartite graph. As a result, firm effect a_i is conditionally independent of all other effects given the effects of its directly connected managers, and symmetrically for manager effects.

The GMRF structure implies that correlation weakens with graph distance. For two nodes at distance d in the bipartite graph,

$$\text{Corr}(x_k, x_\ell) \propto \rho^d. \quad [3]$$

On tree subgraphs, where there is a unique path between any two nodes, this relationship is exact. On graphs with cycles, alternative paths add further contributions to unconditional correlations. This distinction matters for identification by simple covariance ratios, but not for likelihood-based estimation, which uses the full network structure.

From latent effects to observed outcomes. Equation Eq. (1) links the latent effects $\mathbf{x}$ to the observed outcomes $\mathbf{y}$. In matrix form,

$$\mathbf{y} = \mathbf{V}\mathbf{x} + \varepsilon, \quad [4]$$

where $\mathbf{V}$ is a $K \times (N_f + N_m)$ measurement matrix with $V_{k,i} = 1$ if spell k involves firm i and $V_{k,m} = 1$ if spell k involves manager m (all other entries zero), and $\varepsilon \sim \mathcal{N}(\mathbf{0}, \sigma_\varepsilon^2 \mathbf{I}_K)$.

The variance-covariance matrix of observed outcomes is

$$\mathbf{\Omega} = \text{Cov}(\mathbf{y}) = \mathbf{V}\mathbf{Q}^{-1}\mathbf{V}^\top + \sigma_\varepsilon^2 \mathbf{I}_K. \quad [5]$$

This is the sum of the projected latent covariance and the noise variance. The matrix $\mathbf{\Omega}$ is $K \times K$ and dense, so direct inversion is infeasible in large samples.

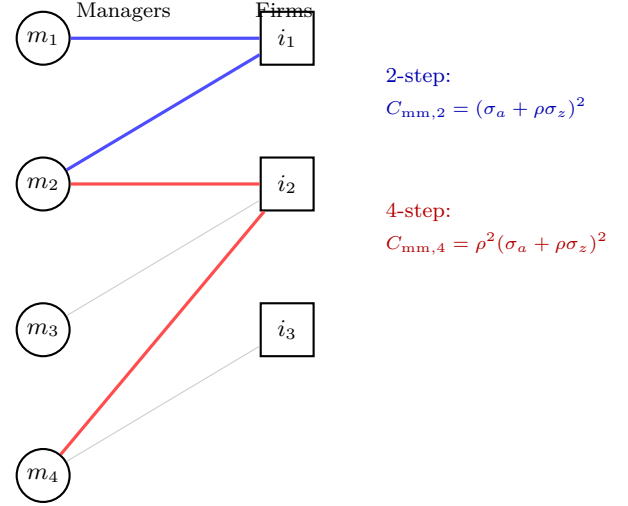


Fig. 1. Identification from network paths. Managers (circles) connect to firms (squares). Blue: 2-step path (m_1 - i_1 - m_2). Red: 4-step path (m_1 - i_1 - m_2 - i_2 - m_4). The ratio $C_{mm,4}/C_{mm,2} = \rho^2$ identifies sorting.

Intuition: correlation decay and network distance. The decay property in Eq. (3) also gives a simple intuition for identification. Consider two managers m_1 and m_2 who worked at the same firm i . They are at distance 2 in the bipartite graph, along the path m_1 - i - m_2 . Their outcome covariance is

$$\text{Cov}(y_{im_1}, y_{im_2}) = (\sigma_a + \rho\sigma_z)^2 \equiv C_{mm,2}. \quad [6]$$

Now consider two managers at distance 4, connected through two firms and an intermediate manager. Their covariance is

$$\text{Cov}(y_{i_1 m_1}, y_{i_2 m_4}) = \rho^2 \cdot C_{mm,2}. \quad [7]$$

The ratio of 4-step to 2-step covariances is therefore ρ^2 , regardless of the variance parameters. This is the path-attenuation idea: each additional pair of hops through the network scales covariance by another factor of ρ^2 .

The same logic extends to longer even-length paths:

$$C_{mm,h} = \rho^{h-2} \cdot C_{mm,2}, \quad [8]$$

An analogous expression holds for firm-firm covariances. Figure 1 illustrates the 2-step and 4-step cases on a small bipartite graph.

Identification and estimation

Maximum likelihood estimation. The log-likelihood of the observed data under the GMRF model is

$$\ell(\theta) = -\frac{K}{2} \ln(2\pi) - \frac{1}{2} \ln \det(\mathbf{\Omega}) - \frac{1}{2} \mathbf{y}^\top \mathbf{\Omega}^{-1} \mathbf{y}, \quad [9]$$

where $\theta = (\sigma_a, \sigma_z, \sigma_\varepsilon, \rho)$ and $\mathbf{\Omega} = \mathbf{V}\mathbf{Q}^{-1}\mathbf{V}^\top + \sigma_\varepsilon^2 \mathbf{I}$. Direct evaluation requires computing $\mathbf{\Omega}^{-1}$ and $\ln \det(\mathbf{\Omega})$, both of which involve a $K \times K$ dense matrix.

The Woodbury matrix identity rewrites the likelihood in terms of a posterior precision matrix, $\mathbf{P} = \mathbf{Q} + \sigma_\varepsilon^{-2} \mathbf{V}^\top \mathbf{V}$, of dimension $(N_f + N_m) \times (N_f + N_m)$. Because $\mathbf{V}^\top \mathbf{V}$ is diagonal, $\mathbf{P}$ inherits the sparsity of $\mathbf{Q}$. This is what makes maximum likelihood feasible at our scale: both the log-determinant and the quadratic form in Eq. (9) can be computed with sparse

Cholesky methods rather than by inverting a dense $K \times K$ covariance matrix. For very large networks, where Cholesky fill-in becomes costly, we approximate the log-determinant with the Hutchinson trace estimator (14) and stochastic Lanczos quadrature, which rely only on sparse matrix-vector products and scale linearly in the number of edges.

Moment conditions and informal identification. Before turning to the full likelihood, it is useful to summarize the model’s moment implications. The model predicts five key moments:

1. Total variance: $V = \sigma_a^2 + \sigma_z^2 + 2\rho\sigma_a\sigma_z + \sigma_\varepsilon^2$.
2. Manager–manager 2-step covariance: $C_{mm,2} = (\sigma_a + \rho\sigma_z)^2$.
3. Firm–firm 2-step covariance: $C_{ff,2} = (\sigma_z + \rho\sigma_a)^2$.
4. Manager–manager 4-step covariance: $C_{mm,4} = \rho^2 \cdot C_{mm,2}$.
5. Firm–firm 4-step covariance: $C_{ff,4} = \rho^2 \cdot C_{ff,2}$.

These five conditions identify four parameters with one over-identifying restriction. In particular, the ratio structure $\rho^2 = C_{mm,4}/C_{mm,2} = C_{ff,4}/C_{ff,2}$ isolates the sorting parameter. Higher-order moments, such as 6-step and 8-step covariances, provide additional specification checks.

On trees (acyclic subgraphs), these moment conditions are exact because there is a unique path between any two nodes. On graphs with cycles, multiple paths between the same pair of nodes create additional contributions to covariances, and the simple ratio relationship $\rho^2 = C_4/C_2$ holds only approximately. The full MLE accounts for the complete network topology, including cycles, by working with the exact GMRF likelihood rather than selected moment conditions.

Advantages over fixed effects. Relative to fixed effects, the random-effects approach has three advantages in sparse networks. First, it avoids the limited-mobility bias that arises when many individual effects must be estimated from short careers. Second, it uses the full network rather than focusing only on a connected subset of observations. Third, likelihood-based estimation exploits the entire covariance structure implied by the graph, not just selected moments.

Data and network structure

Hungarian CEO–firm panel. We use administrative data covering the universe of Hungarian firms and their registered CEOs from 1990 to 2018. The data combine a firm registry maintained by corporate courts with annual balance-sheet filings, covering more than one million firms and 1.3 million managers linked through 1.9 million employment spells (see Methods for data sources, entity resolution, and sample construction).

Network descriptive statistics. Table 1 summarizes the bipartite CEO–firm graph.

The degree distribution is highly skewed: most CEOs manage only one or two firms over the sample period (Figure 2, panel a). About 75% of managers are observed at only one firm, and 49% of firms have only one CEO over the full sample period. The upper tail is substantial, however: some managers run dozens of firms, and some firms cycle through many CEOs.

The network decomposes into about 511,000 connected components. The largest connected component contains roughly

378,000 managers and 337,000 firms, linked by 809,000 edges—about 42% of all edges. The remaining nodes belong to small components, many of which are isolated firm–manager pairs (Figure 2, panel b). This fragmentation reflects the low mobility of CEOs over the sample period.

Within the largest connected component, the typical shortest path between two randomly selected managers is about 20 edges (Figure 2, panels c and d). This feature foreshadows the fixed-effects problem: comparisons often depend on long chains of intermediate links, so estimation noise accumulates rapidly.

Table 1. Descriptive statistics of the manager–firm mobility network.

	Value
<i>Full network</i>	
Managers	1,290,613
Firms	1,029,931
Manager–firm spells	1,918,820
Connected components	511,244
<i>Giant component</i>	
Managers	377,572
Firms	337,215
Edges	808,798
Share of managers	29.3%
Share of firms	32.7%
<i>Path lengths (giant component)</i>	
Mean, manager pairs	19.6
Median, manager pairs	20
Mean, firm pairs	19.0
Median, firm pairs	18

Results

Main estimates. Table 2 reports the GMRF maximum-likelihood estimates for the pooled sample period (1990–2018). Our baseline specification restricts the network to components with at least 2 firms and 2 managers, excluding isolated pairs that contain no information about sorting. The estimate is $\hat{\rho} = 0.706$, implying strong positive assortative matching: better managers tend to work at better firms. Firm heterogeneity is much larger than manager heterogeneity ($\hat{\sigma}_a = 1.41$ versus $\hat{\sigma}_z = 0.19$), while the match-specific noise is $\hat{\sigma}_\varepsilon = 1.20$.

The low $\hat{\sigma}_z$ does not mean that managers are unimportant. With $\rho \approx 0.7$, much of their contribution operates through the covariance between firm and manager effects rather than through manager variance in isolation.

Table 2. GMRF maximum likelihood estimates for the pooled sample period (1990–2018). Outcome is log real revenue, year-demeaned. Baseline sample restricts to components with at least 2 firms and 2 managers.

	$\hat{\rho}$	$\hat{\sigma}_a$	$\hat{\sigma}_z$	$\hat{\sigma}_\varepsilon$
Pooled, $N_f \geq 2, N_m \geq 2$	0.706	1.413	0.187	1.196

Notes: $N_f=530,213$ firms, $N_m=617,613$ managers, $K=1,131,996$ edges. Estimation uses the Hutchinson/SLQ approximation with Nelder-Mead optimization.

Network component analysis. A key concern is whether the results are driven by small, weakly connected components that

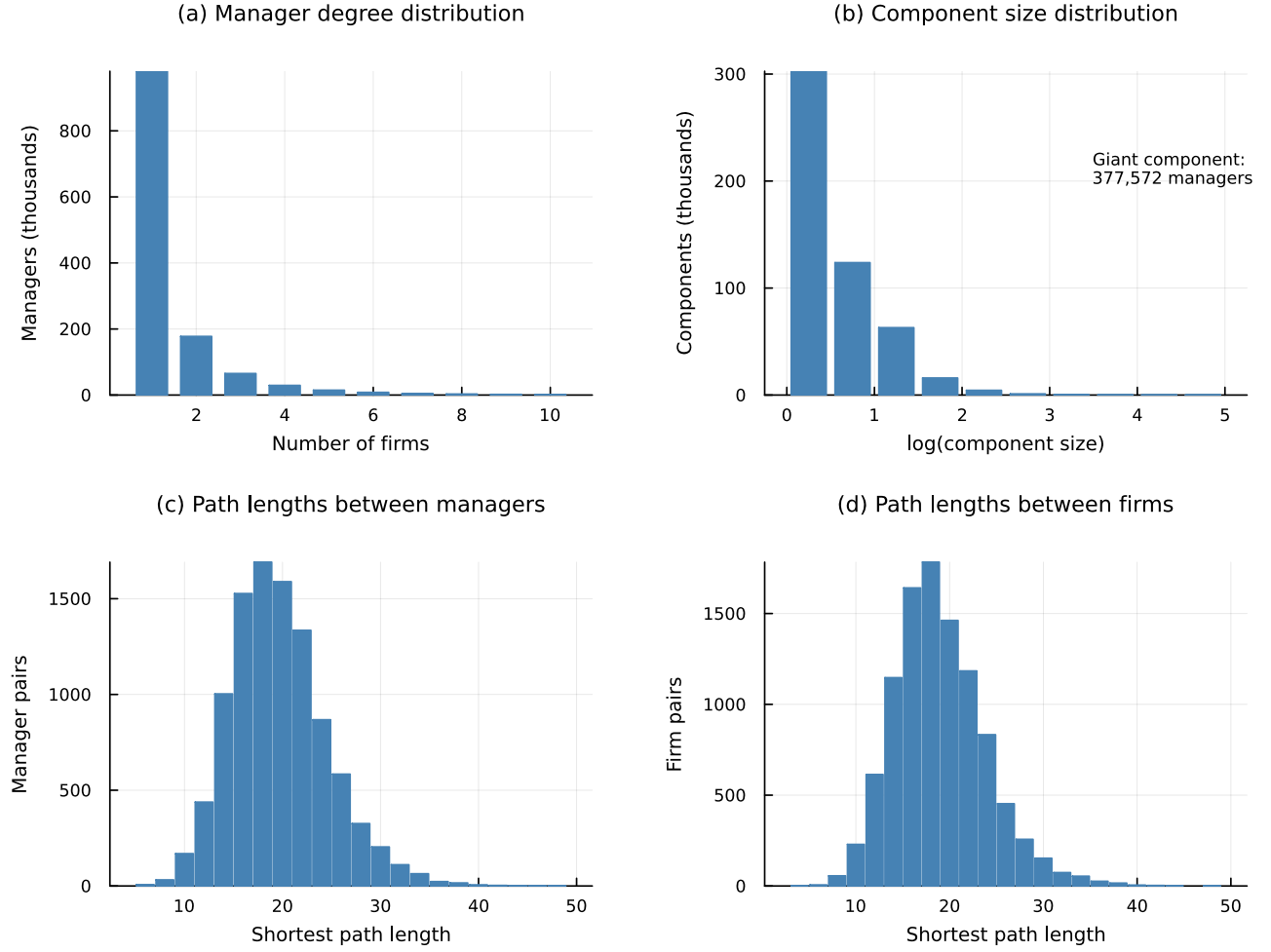


Fig. 2. Structure of the bipartite CEO–firm mobility network, 1990–2018. Panel (a): degree distribution of managers (number of distinct firms managed). Panel (b): component size distribution (by number of managers); the giant component with 377,572 managers is not shown. Panels (c) and (d): distribution of shortest path lengths between 10,000 randomly sampled pairs of managers and firms, respectively, within the giant component.

contain little information about sorting. To assess this, we re-estimate the model on subnetworks defined by progressively stricter connectivity requirements: the full network, components with at least $K \geq 2, 3, 5$ edges, components with at least 2 firms and 2 managers, and the giant component.

Table 3 reports results for the pooled sample period. The sorting estimate $\hat{\rho}$ increases monotonically with the minimum component-size cutoff: from 0.52 on the full network to 0.62 when we keep components with at least 2 edges, 0.71 in the baseline sample, and 0.75 on the giant component.

The monotonic increase in $\hat{\rho}$ is consistent with small components pulling the estimate toward zero. Components with only 2 or 3 edges contain little information about the covariance structure needed to identify ρ , whereas larger components expose richer path structure and therefore sharper information about sorting.

The firm effect parameter $\hat{\sigma}_a$ is stable across component restrictions, at about 1.4 to 1.5, suggesting that firm heterogeneity is robust to sample composition. The manager effect parameter $\hat{\sigma}_z$ varies more across specifications but remains small relative to $\hat{\sigma}_a$. The match-specific noise $\hat{\sigma}_\varepsilon$ increases somewhat with stricter component restrictions, consistent with

denser subnetworks having more variable outcomes.

Table 3. GMRF estimates by minimum component size, pooled sample (1990–2018). The sorting parameter $\hat{\rho}$ increases monotonically with minimum component size.

Sample	Edges	$\hat{\rho}$	$\hat{\sigma}_a$	$\hat{\sigma}_z$
Full network	1,730,354	0.516	1.477	0.236
$K \geq 2$	1,483,394	0.624	1.442	0.210
$K \geq 3$	1,271,609	0.680	1.428	0.194
$N_f \geq 2, N_m \geq 2$	1,131,996	0.706	1.413	0.187
$K \geq 5$	1,057,178	0.728	1.427	0.184
Giant component	733,112	0.752	1.475	0.193

Notes: K is the number of edges (manager–firm spells). “Full network” includes all components; “ $K \geq k$ ” restricts to components with at least k edges; “ $N_f \geq 2, N_m \geq 2$ ” restricts to components with at least 2 firms and 2 managers (baseline specification, in bold); “Giant component” restricts to the largest connected component. $\hat{\sigma}_\varepsilon$ ranges from 1.12 (full) to 1.26 (giant). All estimates use the Hutchinson/SLQ approximation with Nelder-Mead optimization.

Variance decomposition. Table 4 reports the variance decomposition of log revenue under the estimated GMRF model for the pooled period. Total variance decomposes into four additive components:

$$V_{\text{total}} = V_{\text{firm}} + V_{\text{manager}} + V_{\text{cross}} + V_{\varepsilon},$$

where $V_{\text{cross}} = 2\rho\sigma_a\sigma_z$ (network-weighted covariance) captures the contribution of positive assortative matching.

These variance components are not simply the parameter values σ_a^2 , σ_z^2 , and σ_{ε}^2 . Because the latent effects are linked through the network, the parameters describe conditional variation, whereas the observed variance of outcomes reflects the unconditional covariance structure implied by the full graph and is computed from Ω .

Firm effects dominate, accounting for 62.3% of total revenue variance ($V_{\text{firm}} = 2.92$). Manager effects are small in isolation (1.0%, $V_{\text{manager}} = 0.05$), but the covariance between firm and manager effects contributes another 6.1% ($V_{\text{cross}} = 0.29$). Match-specific noise accounts for the remaining 30.5% ($V_{\varepsilon} = 1.43$).

The 6.1% covariance share is the direct contribution of positive assortative matching to revenue dispersion in the model. In a counterfactual that raises ρ to 1 (perfect sorting), the covariance term increases by a factor of $1/\hat{\rho} \approx 1.4$. Under a log-normal approximation, this translates into an aggregate output gain of about 6%.

Table 4. Variance decomposition of log revenue, pooled period (1990–2018), components with $N_f \geq 2$ firms and $N_m \geq 2$ managers.

Component	Variance	Share
Firm effects (V_{firm})	2.923	62.3%
Manager effects (V_{mgr})	0.049	1.0%
Cross-covariance (V_{cross})	0.286	6.1%
Match noise (V_{ε})	1.432	30.5%
Total (V_{total})	4.690	100.0%

Comparison with two-way fixed effects. To benchmark our results, we estimate a standard two-way fixed-effects model on the giant component of the pooled network. The AKM regression yields $\hat{\rho}_{\text{TWFE}} = -0.64$ —strongly negative and opposite in sign to our GMRF estimate of $+0.75$ on the same sample. The leave-one-out bias correction of Kline, Saggio, and Sølvesten (15) reduces the magnitude but does not resolve the discrepancy: the corrected estimate is $\hat{\rho}_{\text{KSS}} = -0.39$, still negative.

This dramatic reversal is consistent with limited mobility bias. In the CEO network, the giant component has a mean path length of about 20, so comparing two firms typically requires many intermediate links. Each link adds estimation error to the fixed-effect estimates. The accumulated noise inflates the estimated variance of both firm and manager effects and can push their correlation toward zero or negative values. The random-effects approach avoids this problem by estimating distributional parameters rather than individual effects.

The failure of the leave-one-out correction in this setting is notable. Although KSS provides consistent variance-component estimates in dense networks, its effectiveness depends on the leave-one-out exercise removing only a limited

share of the information about each individual effect. In the extremely sparse CEO network, where most managers are observed at only one firm, omitting a single observation may remove nearly all information about a manager’s effect, making the correction too fragile.

Counterfactual analysis

The output effect of sorting. We quantify the aggregate output consequences of CEO–firm sorting. Under the additive model $y_{im} = a_i + z_m + \varepsilon_{im}$, and under a log-normal approximation for output, expected output is

$$\begin{aligned} \mathbb{E}[Y] &= \mathbb{E}[\exp(y)] = \exp\left(\mu + \frac{1}{2}\text{Var}(y)\right) \\ &= \exp\left(\mu + \frac{1}{2}(V_{\text{firm}} + V_{\text{manager}} + V_{\text{cross}} + V_{\varepsilon})\right), \end{aligned}$$

where $\mu = \mathbb{E}y$ is mean log output. An increase in ρ , holding the remaining variance components fixed, raises the variance of log output and therefore raises expected output through Jensen’s inequality.

Using our baseline estimates, we ask how much output would increase if sorting were perfect ($\rho = 1$). The covariance term scales with ρ , so perfect sorting raises V_{cross} by a factor of $1/\hat{\rho} = 1.42$. The additional covariance is

$$\Delta V_{\text{cross}} = V_{\text{cross}} \times (1/\hat{\rho} - 1) = 0.286 \times 0.42 = 0.12.$$

Under the log-normal approximation, this translates into an output gain of $\frac{1}{2}\Delta V_{\text{cross}} \approx 0.06$ log points, or about 6%. Perfect sorting would raise aggregate output by about 6% relative to the current allocation.

Is ρ a structural parameter? The counterfactual is informative if ρ is a structural parameter—that is, if it is invariant to the policies or shocks being considered. We do not claim this definitively. In an equilibrium matching model with search frictions, ρ arises endogenously from the interaction of preferences, technologies, and search frictions. Changing the matching technology would change ρ but also potentially change the network structure and the variance parameters.

We take the view that ρ is a reduced-form summary of matching efficiency that is useful for accounting purposes, even if it is not a deep structural parameter. The counterfactual answers the question: how much output variation is attributable to sorting, holding other features of the economy fixed? This is analogous to growth accounting exercises that decompose output changes into contributions from capital, labor, and TFP without claiming that TFP is a structural parameter.

Discussion

Our GMRF random-effects model finds strong positive CEO–firm sorting ($\rho \approx 0.7$ – 0.75) in Hungarian administrative data spanning three decades. This contrasts with the negative correlation ($\rho \approx -0.6$) implied by two-way fixed effects on the same data, which we attribute to limited mobility bias in an extremely sparse CEO network.

Several extensions merit future work. Observable characteristics can be incorporated either by residualizing outcomes or by joint maximum likelihood estimation (see Methods). Empirical Bayes posterior means for individual effects can be

computed from the estimated GMRF, yielding shrinkage estimates that are more stable than raw fixed effects. Monte Carlo exercises would clarify finite-sample performance across different network structures. The approach is readily applicable to other bipartite mobility settings, including worker–firm wage data, student–school achievement data, and patient–provider health outcomes. Embedding the GMRF in an equilibrium assignment model would provide firmer foundations for policy counterfactuals.

Materials and Methods

Observables. Observable characteristics can be included in the model in a straightforward manner. Consider the extended specification

$$y_{im} = \mathbf{x}_{im}^\top \beta + a_i + z_m + \varepsilon_{im},$$

where $\mathbf{x}_{im}$ is a vector of observables, such as industry, region, or year effects. There are two estimation approaches. The first is a two-step procedure: estimate β by fixed-effects regression, residualize $\tilde{y}_{im} = y_{im} - \mathbf{x}_{im}^\top \beta$, and apply the GMRF estimator to $\tilde{y}$. The second is joint maximum likelihood estimation, which profiles β out of the Gaussian likelihood together with the GMRF parameters. Both approaches are computationally feasible because adding observables shifts the mean of the Gaussian model without altering its covariance structure.

Network formation. Our model takes the network as given and estimates the distribution of latent effects conditional on the observed matching pattern. The network itself is an endogenous outcome of the labor market—which managers work at which firms is determined by search, matching, and separation processes. We do not model network formation. A natural extension would embed the GMRF in an equilibrium assignment model where match probabilities depend on complementarities between firm and manager types.

Data sources and construction. We combine two administrative datasets covering Hungarian firms from 1990–2018. The firm registry, maintained by corporate courts, tracks all registered executives authorized to represent companies, including CEOs and other officers with signatory authority. Each record spans a time interval, updating when positions change or identifiers are modified. The registry contains names, addresses, birthdates (from 2010), and mother’s names (from 1999), with numerical identifiers available only from 2013 onward. The balance sheet dataset provides annual financial reports for all firms with double-entry bookkeeping, including revenue, employment, sector, and ownership structure. Together, these sources cover over one million firms.

Tracking individual CEOs across firms and time requires entity resolution. From 2013 forward, numerical identifiers enable direct linking. For earlier periods, we match records using names, addresses, mother’s names, and birthdates to create unique person identifiers. Matching quality improves substantially after 1999 (when mother’s name reporting begins) and 2010 (when birthdate reporting starts), though the 1990s data achieves reasonable coverage through name and address combinations.

Identifying CEOs among registered representatives uses several heuristics. When explicit managing director titles exist, we apply them directly. For firms with three or fewer representatives, we assume all are CEOs. When more than three representatives are present, we assign CEO status based on continuity with previously identified CEOs at the same firm.

We apply five sample restrictions. First, we exclude mining and finance sectors due to specialized accounting frameworks. Second, we drop firms ever having more than 2 simultaneous CEOs to avoid complex governance structures. Third, we remove firms with more than 10 CEO changes over the sample period to reduce noise from classification errors. Fourth, we exclude all state-owned enterprises, as their objectives differ from private businesses. Fifth, we exclude public firms, joint-stock companies, and cooperatives where governance structures differ from standard private corporations. We

retain all private firms including nonemployer businesses, as single-person companies represent a substantial portion of the Hungarian corporate sector.

The outcome y_{im} is log real firm revenue (deflated using standard price indices) aggregated over each time window while manager m is in office at firm i . If a firm–manager relation spans multiple years within a window, we average log revenues across years.

Data availability. We use administrative records on Hungarian firm managers and firm financial statements. Individual court records are public, but the compiled dataset cannot be shared. We purchased the raw administrative data from Opten Zrt., an authorized redistributor of Hungarian corporate court records, and the licensing agreement does not permit redistribution of the original raw files. Upon publication, we will share all data-cleaning and analysis code needed to reproduce the results from the vendor data and will provide access details in the Data Availability and Provenance Statement.

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Significance Statement

How managers and firms match matters for allocative efficiency. We develop a random effects model for bipartite mobility networks that estimates sorting from firm revenues and CEO career histories. The model replaces hundreds of thousands of individual effects with four distributional parameters, avoiding the limited mobility bias that biases standard fixed-effects estimates. The associated precision matrix is sparse, making likelihood-based estimation feasible on very large networks. Applied to three decades of Hungarian data, the method reveals strong positive assortative matching (correlation ≈ 0.7), whereas standard fixed-effects methods imply negative sorting.

M.K. conceived the idea; M.K. and K.O. designed the study; M.K., Á.T., and U.W. prepared the data; U.W. developed computational methods and performed data analysis; M.K., K.O., Á.T., and U.W. wrote the paper.

The authors declare no competing interests.

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